

Cambodian Cancer Incidence Rates in California and Washington, 1998–2002

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BACKGROUND. Among the many diverse Asian ethnic groups living in the US, Cambodian immigrants comprise a small fraction (1.8%) of the total Asian population. Because of their small numbers, Cambodian vital statistics are often combined into Southeast Asian (SA) cancer data consisting of Vietnamese, Thais, Laotians, and Hmong.

METHODS. The 2000 Census counts were used for 2 Cambodian populations, Cambodians alone and Cambodians alone and in combination with any other racial/ethnic group for California and for Seattle (Puget Sound area), Washington. Then the cancer incidence rates were calculated using cancer cases from the California and Puget Sound cancer registries between 1998–2002. The 1998–2002 annual age-adjusted incidence rates, upper bound rates (based on the Cambodian alone population), lower bound rates (based on the Cambodians alone or in combination population) are reported and compared with the rates in the non-Hispanic White (NHW) population in these regions.

RESULTS. The top 5 cancers in Cambodian males are lung and bronchus, liver, prostate, colorectal, and stomach cancers. The sites where the rates are higher in male Cambodians than NHW males are (in ascending rank) nasopharynx, liver, stomach, myeloma, and lung and bronchus. The top 5 cancers for female Cambodians are breast, lung, colon and rectum, cervix, and thyroid. The sites where female rates are greater than NHW female rates are (in ascending rank) nasopharynx, liver, stomach, cervix uteri, oral cavity, and thyroid.

CONCLUSIONS. The challenges to address the health issues of Cambodians are complicated by historical events that caused their emigration to the US. Many of the immigrants are survivors of the holocaust in Cambodia. Health programs for Cambodians must deal with the consequences of these issues as well as cultural issues of language and religion in helping Cambodians to reduce their cancer disparities. *Cancer* 2007;110:1370–5. © 2007 American Cancer Society.

KEYWORDS: Cambodians, cancer incidence.

In 1975, Pol Pot and the Khmer Rouge communist regime came to power in Cambodia. Their reign lasted less than 5 years, but it is estimated that 1.5 to 3 million of the 7 million population were killed.¹ The 1984 movie *The Killing Fields* depicted these events. From 1983 to 1985, a large number of Cambodian refugees emigrated to the US.² As 1 of the more recent Asian immigrant groups, Cambodian refugees make up only 1.8% of the total Asian populations.³ The majority resettled in California, Washington, Texas, Pennsylvania, and Massachusetts, where they can either be reunited with their families and friends and/or where US refugee programs were available to assist them.^{2–4}

There are few published studies on the incidence rates of cancer among the Cambodian population in the US. These articles mainly

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discuss female Cambodian cancer issues. Taylor et al.⁵ wrote several articles on cervical cancer control and intervention programs. Tu et al.⁶ focused on breast cancer, and Mahloch et al.⁷ provided input on cultural bridging in the development of educational materials on cancer. Several factors contribute to the lack of published studies. One is that it is a small population. Cambodian cancer data are usually combined with other groups in the Southeast Asian (SA) area that include Vietnamese, Thai, Laotians, and Hmong.⁸ Thus, it is difficult to interpret the research findings for the specific subgroups.^{8,9} Cultural sensitivities and language barriers also play important roles in the limited data of Cambodian health status.^{2,4} Traditional medicine, such as rubbing the body with a coin to relieve body ache or symptoms, is a common practice in Cambodian households.³ Furthermore, the lack of physician representation and gender match between physician and patient are additional obstacles to access of healthcare treatment.^{2,9} In all, the major challenge is communication, as some Cambodians are illiterate in their native language and speak little or no English.^{2,9,10}

Cancer incidence rates in Cambodian immigrants can play an important role for healthcare providers and policymakers. Knowing their leading cancers can allow special attention to these cancers, which differ from the common cancers in the non-Hispanic White (NHW) population. Our study examines the cancer incidence rates from the Surveillance, Epidemiology, and End Results (SEER) Program cancer registries for the state of California and Puget Sound, Washington for the years 1998 to 2002. We obtained population denominators for rates from the 2000 Census, which reports 2 population counts as *alone*, i.e., those identifying themselves as being of only of Cambodian ethnicity (forming the upper boundary rate), and as *alone and/or in combination*, i.e., those identifying themselves as being of Cambodian ethnicity only and/or in combination with other racial/ethnic groups (lower boundary rate).

MATERIALS AND METHODS

The number and type of cancer cases for many Asian ethnicities, such as Cambodians, are routinely collected by SEER cancer registries since 1988. In our study we used cancer cases for Cambodians diagnosed from 1998–2002 for the state of California and the SEER Puget Sound cancer registry for the state of Washington. A total of 581 cases of cancers were diagnosed in California and Puget Sound, Washington between 1998 and 2002, including 308 male cases and 273 female cases. However, the age-specific popula-

tions needed to calculate age-adjusted cancer rates are generally only available from decennial Census data. We used the 2000 Census data to determine the age-specific populations for Cambodians. Then we multiplied the 2000 Census age-specific population counts by 5 to cover the 1998–2002 years of cancer cases.

The cancer cases from California and Puget Sound cancer registries represent 94.5% of all the cancer cases for Cambodians in the 13 SEER registries. Data on cancer sites with 3 or more cases were reported. Thus, the data from these 2 registries represent the best estimate of their rates. All rates are age-adjusted to the US 2000 standard population.

The cancer incidence rates for NHW and Asian and Pacific Islanders for the California and Puget Sound cancer registries were also determined for the period 1998–2002. The ratio of the Cambodian age-adjusted rate to the age-adjusted rate for NHWs for the same sex was determined for each cancer site using the upper bound estimate of the rate. The rank order of the ratios where the Cambodians had higher rates than NHWs was also calculated.

RESULTS

Table 1 summarizes the 1998–2002 annual average age-adjusted rates with an upper bound rate (using the Cambodian alone population) and a lower bound rate (using the Cambodians alone or in combination population) and the number of cancer cases for each cancer site. The incidence rates for NHW and Asians and Pacific Islanders (API) are also reported. The ratio of the Cambodian rate to the NHW rate and the rank order of the ratio if it is greater than 1 are also reported.

The 5 leading cancers in Cambodian males are lung and bronchus, liver, prostate, colorectal, and stomach cancers. The sites where the rates are higher in male Cambodians than NHW males are (in ascending rank) nasopharynx, liver, stomach, myeloma, and lung and bronchus. The top 5 cancer sites for female Cambodians are breast, lung and bronchus, colorectal, cervix, and thyroid. The sites where female rates are greater than NHW female rates are (in ascending rank) nasopharynx, liver, stomach, cervix, and oral cavity.

Male APIs have the same top 5 cancers as the Cambodians, but the order has changed. The top 5 cancers for male APIs are prostate, lung and bronchus, colorectal, liver, and stomach cancers. Female APIs have slightly different leading cancers with breast, colon and rectum, lung and bronchus, corpus and uterus, and thyroid cancers. Cervical cancer is not in the top 5 for API females.

TABLE 1
1998–2002 Cambodian Cancer Incidence Rates (for Sites With ≥3 Cancer Cases) in California and Puget Sound Cancer Registries

Cancer site	High rate	Low rate	Count	Non-Hispanic white rate	Asian and Pacific Islanders rate	Cambodian/White high rate ratio	Rank of ratio
Men							
All sites	386.3	328.1	308	569.1	358.5	0.68	
1 Lung and bronchus	87.1	73.2	56	81.1	57.2	1.07	5
2 Liver and intrahepatic bile duct	51.0	43.2	57	6.8	24.0	7.50	2
3 Prostate	40.2	34.1	25	163.1	91.6	0.25	
4 Colon and rectum	29.1	24.9	27	60.6	49.4	0.48	
5 Stomach	25.2	21.8	16	9.7	19.6	2.60	3
6 Non-Hodgkin lymphoma	23.1	19.4	22	25.0	16.6	0.92	
7 Oral cavity and pharynx	16.5	13.9	14	17.1	6.1	0.96	
8 Leukemia	14.9	12.4	11	16.8	9.3	0.89	
9 Urinary bladder	14.4	12.2	9	42.3	16.4	0.34	
10 Pancreas	9.5	8.3	8	12.8	9.3	0.74	
11 Kidney and renal pelvis	9.4	7.9	5	16.1	7.6	0.58	
12 Nasopharynx	8.6	7.4	7	0.6	4.8	14.33	1
13 Myeloma	7.8	6.7	6	6.3	3.8	1.24	4
14 Brain and other nervous system	6.1	5.2	5	9.4	3.9	0.65	
15 Thyroid	3.2	2.8	4	4.2	3.7	0.76	
Women							
All sites	223.3	189.8	273	457.2	286.5	0.49	
1 Breast	41.0	34.8	58	155.5	87.7	0.26	
2 Lung and bronchus	26.2	22.4	26	59.7	27.2	0.44	
3 Colon and rectum	22.5	19.1	25	44.4	37.1	0.51	
4 Cervix uteri	15.0	12.7	20	7.7	9.2	1.95	4
5 Thyroid	14.4	12.1	20	11.1	11.9	1.30	6
6 Liver and intrahepatic bile duct	14.2	12.1	15	2.7	8.9	5.26	2
7 Non-Hodgkin lymphoma	11.8	10.1	14	17.0	11.1	0.69	
8 Oral cavity and pharynx	10.8	9.2	11	7.0	3.7	1.54	5
9 Stomach	10.3	8.7	12	4.1	11.3	2.51	3
10 Leukemia	9.1	7.7	13	9.7	5.9	0.94	
11 Corpus and uterus, NOS	6.0	5.1	8	25.6	15.3	0.23	
12 Ovary	6.0	5.1	9	15.8	9.9	0.38	
13 Pancreas	4.0	3.5	3	9.8	7.7	0.41	
14 Brain and other nervous system	3.8	3.2	5	6.6	2.8	0.58	
15 Nasopharynx	2.7	2.2	3	0.2	2.0	13.50	1
16 Melanoma of the skin	2.1	1.7	4	21.0	1.2	0.10	

NOS indicates not otherwise specified.

DISCUSSION

Our study indicates the important cancers for Cambodians in the US. We now discuss these cancers in terms of their global distribution, etiology, and some cancer prevention and control activities associated with the top cancers for Cambodians.

Nasopharynx Cancer

Nasopharynx cancer is a relatively rare malignancy in most countries; however, peoples from southeast Asia, China, northeast India, North Africa, and Inuits (Eskimos) of Canada have higher risks.^{11–13} Males are more likely to have the disease than females with an M:F ratio of 2.3:1 in the general population.¹¹ Our incidence rates had an M:F ratio of 3.2:1 in favor of males. Common causes of this malignancy include environment,

such as dust and smoke, and lifestyle, with consumption of salted and preserved foods.¹¹ Epstein-Barr virus (EBV) is well-documented in nasopharyngeal carcinoma. Its viral influence and impact on the predisposition to cancer have yet to be elucidated.¹² In our study, nasopharyngeal cancer was the twelfth leading cancer for Cambodian men (8.6/100,000, 14 times greater than the rate for male NHW rate) and the fifteenth cause for Cambodian women (2.7/100,000, 14 times greater than the female NHW rate).

Liver Cancer

Morbidity and mortality rates in liver cancer are high in developing countries, specifically in sub-Saharan Africa, eastern and southeastern Asia, and Melanesia.¹¹ The 2 major viruses contributing to the global

risks for liver cancer are infections with hepatitis B and C. Development of hepatitis B immunization helps prevent infection in childhood, and has drastically reduced the incidence of liver cancer.¹¹ Other reported risk factors found among Southeast immigrants include *Mycobacterium tuberculosis* (TB) and some intestinal parasites.¹⁴ In our study the liver cancer incidence is the second leading cancer for Cambodian men (51.0/100,000, 7.5 times greater than the male NHW rate) and the sixth leading cancer for women (14.2/100,000, 5.3 times greater than the female NHW rate).

Thyroid

At the global level, thyroid cancer is more prevalent among females in developing countries than among females in the developed countries, 60 cases per 1000 versus 44 cases per 1000, respectively.¹¹ Family history of thyroid cancer, endocrine abnormality, goiter, and iodine or seafood intake are possible factors in the malignancy.^{15,16} Haselkorn et al.¹⁶ proposed that the consumption of cruciferous vegetables, antioxidant vitamins, and phytoestrogens may provide protective effects against thyroid cancer. In our study, Cambodian thyroid cancer was the fifth leading cancer in Cambodian women (14.4/100,000, 1.3 times higher than the rate for female NHWs), whereas it was the lowest incidence for Cambodian men.

Stomach Cancer

Male stomach cancer has some of the highest cancer deaths (700,000 deaths annually) worldwide, but takes fifth place behind lung, liver, prostate, and colon and rectum cancers in our data. The incidence rate seems to be low in men of Southern Asia, North and East Africa, North America, Australia, and New Zealand.¹¹ In our study, the incidence rates for stomach cancer are sixth and fifth among API women and men, respectively. Some contributing factors are consumption of smoked, cured, salted, or pickled foods.¹³ *Helicobacter pylori* infections cannot be overlooked in the etiology of this disease.¹¹ In this study, stomach cancer was the fifth leading cause of cancer for Cambodian men (25.2/100,000, 2.6 times greater than the male NHW rate) and the ninth leading cause of cancer for Cambodian women (10.3/100,000, 2.5 times greater than the rate for female NHWs).

Myeloma

A malignancy of plasma cells in the bone marrow has a higher incidence rate in African-Americans than the White and Asian population, affecting more males than females. Some possible factors contribut-

ing to myeloma are chemicals and occupational exposures, autoimmune disorders, or chronic immune stimulation such as tuberculosis, malaria, and hepatitis.^{11,17} In our study melanoma was the thirteenth leading cause of cancer for Cambodian men (7.8/100,000, 1.2 times greater than male NHW rate) and not reported for Cambodian women.

Lung and Bronchus

Lung cancer is 1 of the leading causes of cancer death among men and women in the US. More than 50% of the lung cancer cases are caused by cigarette smoking.¹⁸ Recently, incidence rates seem to have decreased in men and increased in young women due to an increase in tobacco use observed in that young group.¹⁰ Among young Cambodian females the prevalence of tobacco use is also rising, whereas it is falling among Cambodian males.¹⁰ In this study, lung and bronchus was the top cause of cancer in Cambodian men (87.1/100,000 but 1.1 times the rate for male NHWs) and the second leading cause of cancer for Cambodian women (26.2/100,000, 0.4 times the rate for NHW women).

Oral Cavity and Pharynx

Some studies associate the prevalence of oral cancer with betel quid-chewing by elderly Cambodian women.^{19,20} The chewing involves a combination of areca nut, lime paste, and leaf tobacco that darken the women's teeth who have long chewing habits.²⁰ Some suggest that betel quid-chewing creates new oral flora for new microorganism colonization such as *Candida* species.^{19,20} In our study, oral cancer and pharynx is the seventh leading cause of cancer in Cambodian men (16.5/100,000, the same rate as NHW men) and the eighth leading cause for Cambodian women (10.8/100,000, 1.5 times the rate of NHW females).

Cervix Uteri

The Global Cancer Statistics 2002 reports an estimated 493,000 new cases and 274,000 deaths of cervix cancer among women in the world.¹¹ The incidence and mortality rates are higher in developing countries, specifically in sub-Saharan Africa, Melanesia, Latin America and the Caribbean, southcentral Asia, and southeast Asia.¹¹ Our analysis shows a 95% higher risk in Cambodian women compared with NHW women. A major contribution to the high mortality rate points to insufficient PAP smear tests to detect human papilloma virus (HPV).²¹ Other possible factors include tobacco smoking, high parity, or use of oral contraceptives.²¹ In our study cervical cancer was the fourth leading

cause of cancer in Cambodian women (15.0/100,000, 2 times the rate for NHW females).

Cultural Issues in Addressing Cancer Prevention and Control Activities for Cambodians

Cambodians have unique cultural issues that must be dealt with in developing cancer education programs for this population. Many immigrants from Cambodia were affected by the genocide led by Pol Pot, the leader of the Khmer Rouge (Cambodian Communist Party) from 1975 to 1979. Millions of Cambodians were tortured and executed by the Pol Pot regime. The Pol Pot holocaust left a devastating country of orphans, poverty, famine, and corruption. As survivors from the ravaging war countries, the Cambodian immigrants have been labeled with psychological problems such as depression and post-traumatic stress disorder (PTSD).²² These medical terms are quickly used by medical practices, and the lack of cultural sensitivity for immigrants who experience political repression is often missed.² The Cambodian immigrants do not usually talk about their war experiences during family or social functions, unless one raises the question. In 1 instance, a Cambodian lady who lost her husband and children to Pol Pot's execution blocks the traumatic experience from her memories so that she can cope with her daily life activity. Thus, it is important to realize these issues in working with Cambodians.

This problem is not the only cultural factor that needs to be considered in dealing with the health of Cambodians. Cambodian history dates back to the Angkor era, where King and Buddhism dictated that Cambodian civilization had a fixed social status. Born into royal families or the upper class is a blessing in which power for ruling the kingdom was approved by God. It is unthinkable for a peasant to rise above the status into which he/she was born. This is her/his fate, and they should live prosperously to accumulate good merit for the next life.²³

This culture of fate, acceptance, and endurance of life makes Cambodians stand apart from other societies. It plays a major role in the downfall of Cambodians' politics and their kingdoms throughout their history. Total obedience is the ultimate discipline; silence is the golden rule that not only is taught at an early age but has also been a survival method to escape the punishment inflicted by the Khmer Rouge.² This 'silence' is often misunderstood and misinterpreted by many foreigners as a characteristic of being indifferent, passive, noncompliant, and even not intelligent.²³

To design and implement healthcare awareness for health disparities in this population is a chal-

lenge. It is even more challenging when one must deal with the Cambodian language, culture, and historical events to address these complex issues. The Cambodian immigrants have come a long way, surviving the traumatic experiences under the Pol Pot regime to a new culture that most of them are not accustomed to. They are the 'survivors' of political repression. It is this fundamental survival skill that Western medical teams need to understand and sympathize with, and to provide a healthcare system that is geared to the needs of this special group.

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